Joseph M. Ryan

joseph.ryan@duke.edu Languages: English and French (native), Spanish (working)
Background: Winchester College alumnus

Education

Duke University, Ph.D. Candidate in Physics (Quantum Computing) at the Duke Quantum Center. 2022–present
Advisor: Crystal Noel, Theory Collaborators: Stephen Teitsworth.

University of Michigan, B.S. in Honors Physics & Mathematics, High Honors. 2022
Recipient of William L. Williams Award for thesis on QCD color entanglement (LHCb).

Research Summary

Physicist specializing in **quantum stochastic processes.**

Research Experience

Duke Quantum Center (Noel Lab) 2022–present

- Built and operated a trapped-ion quantum computer

Univ. of Michigan – Aidala Group (LHCb Experiment at CERN) 2021–22

- Designed statistical methods to detect TMD factorization breaking due to color entanglement in proton collisions; analyzed correlated stochastic parton dynamics.

Publications

- J. M. Ryan *et al.*, "Experimental Measurement of Quantum First-Passage-Time Distributions." Physical Review Research 8, L022025, (2026). Featured in *Quantum Zeitgeist*.
- J. M. Ryan *et al.*, "Quantum-classical crossover in noisy monitored oscillators." arXiv:2608.08267v1 (2026).

Selected Talks

- APS DAMOP (2025) – *Noise-Driven Quantum First-Passage-Time Distributions in Ion Traps*
- NSF Robust Quantum Computing, Princeton (2024) – *Quantum Brownian Motion and First-Passage Times.*
- APS March Meeting (2024) – *Noise-Driven Escape Times in Trapped Ion Systems.*

Work Experience

OTC Fellow, Duke Office of Translation & Commercialization 2025-present

- Supporting Duke startups through market analysis and go-to-market strategy development.

Advanced Physics Laboratory, Univ. of Michigan 2020

- Teaching Assistant. Designed and taught advanced atomic, optical and nuclear physics experiments.

Michigan Hyperloop Team 2018–19

- Business director. Raised and managed funds for a multidisciplinary engineering team for hyperloop.

Airbus 2017

- Metrology Intern. Developed precision measurement protocols to ensure quality and accuracy in wing production.

Skills

- **Stochastic Modeling:** Quantum/classical stochastic processes, first-passage times